Data Encryption Standard (DES)

33. Write a C program for Data encryption standard (DES) has been found vulnerable to very powerful attacks and therefore, the popularity of DES has been found slightly on the decline. DES is a block cipher and encrypts data in blocks of size of 64 bits each, which means 64 bits of plain text go as the input to DES, which produces 64 bits of ciphertext. The same algorithm and key are used for encryption and decryption, with minor differences. The key length is 56 bits. Implement in C programming.

Aim:

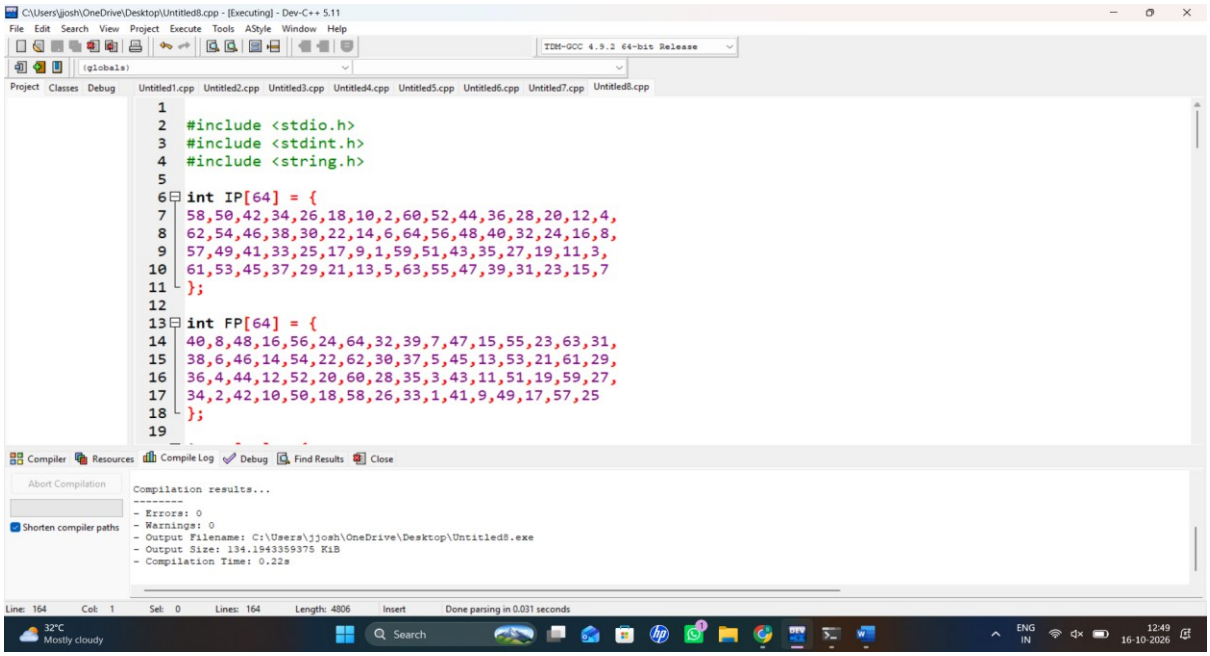
To implement DES encryption and decryption in C using a 64-bit data block and a 56-bit effective key.

Algorithm:

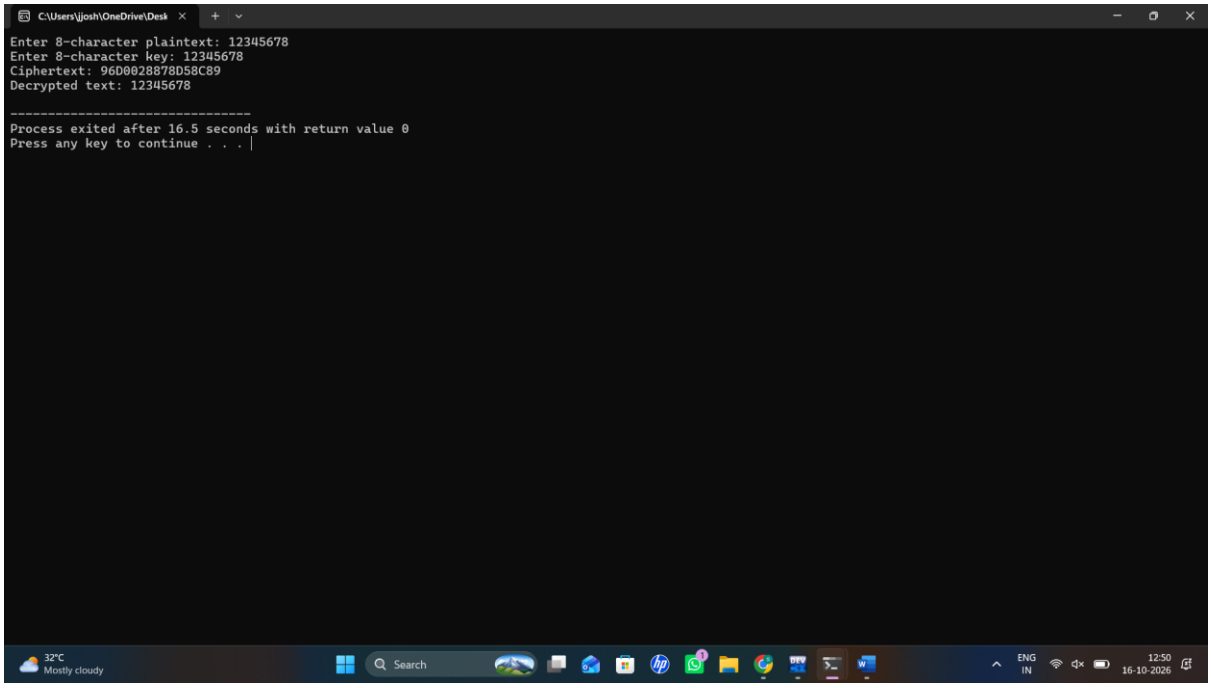
1. Read an 8-byte plaintext and an 8-byte key.
2. Initialize the DES key schedule.
3. Apply DES encryption to the 64-bit plaintext block.
4. Display the ciphertext in hexadecimal format.
5. Decrypt the ciphertext using the same key.
6. Display the recovered plaintext.

Program:

This implementation uses the OpenSSL DES library. DES is obsolete and should only be used for educational purposes.



Output:



Result:

DES encryption and decryption are implemented using an 8-byte block and an 8-byte key representation. DES uses only 56 effective key bits, making it vulnerable to brute-force attacks.

Compilation:

With OpenSSL development files installed, compile using:

```
gcc experiment33.c -o experiment33 -lcrypto
```